

A CERTIFIED GEOMETRIC LOWER BOUND FOR THE THREE-DIMENSIONAL BLASCHKE–LEBESGUE PROBLEM

HYRA

ABSTRACT. We prove that every convex body $K \subset \mathbb{R}^3$ of constant width $w > 0$ satisfies

$$\text{Vol}(K) \geq c_{\text{cert}} w^3, \quad c_{\text{cert}} := \frac{130838246407123}{10^{15}} \pi, \\ c_{\text{cert}} > 0.411040473721188.$$

The proof combines the contact geometry of a minimum circumscribed ball, an opposite-point identity specific to bodies of constant width, a divergence estimate for the radial function, and a tangent-line majorant whose nonnegative deficit is recovered on eight disjoint spherical caps. The last step is a finite computer-assisted verification carried out entirely with exact rational inequalities; floating-point arithmetic is used only to propose candidates that are subsequently rechecked exactly. Independently of the computer-assisted part, the same method yields the explicit analytic bound

$$\text{Vol}(K) \geq \frac{533032}{4143735} \pi w^3 = 0.404120778796972 \dots w^3.$$

The certified bound reaches 97.8994% of the volume of a Meissner tetrahedron of the same width.

1. INTRODUCTION

A convex body $K \subset \mathbb{R}^3$ has *constant width* $w > 0$ if the distance between its two supporting planes orthogonal to any unit vector is w . For $w > 0$, set

$$m_3(w) := \inf\{\text{Vol}(K) : K \subset \mathbb{R}^3 \text{ is a convex body of constant width } w\}.$$

Dilation gives $m_3(w) = w^3 m_3(1)$. The planar minimum-area problem was solved independently by Lebesgue and Blaschke: the minimizers are the Reuleaux triangles [8, 2]. In dimension three, determining $m_3(w)$ remains the classical Blaschke–Lebesgue problem. Bonnesen and Fenchel conjectured that the minimum is attained by the two noncongruent Meissner tetrahedra [4, 9, 7]. Their common volume is

$$V_M(w) = \frac{\pi}{12} \left(8 - 3\sqrt{3} \arccos \frac{1}{3} \right) w^3 = 0.419860045965080 \dots w^3; \quad (1.1)$$

Date: August 15, 2026.

2020 Mathematics Subject Classification. Primary 52A20; Secondary 52A40, 65G30.

Key words and phrases. body of constant width, Blaschke–Lebesgue problem, Meissner conjecture, convex geometry, computer-assisted proof, exact rational arithmetic.

see, for example, [1, 3].

Chakerian proved the lower bound

$$\text{Vol}(K) \geq \frac{\pi}{3}(3\sqrt{6} - 7)w^3 = 0.364916122595 \dots w^3$$

in [5]. A recent spectral argument of Nishioka improved this to

$$\text{Vol}(K) \geq \frac{4\pi}{33}w^3 = 0.380799109526 \dots w^3; \quad (1.2)$$

see [10]. The purpose of this paper is to establish the following further improvement.

Theorem 1.1 (Main theorem). *Every convex body $K \subset \mathbb{R}^3$ of constant width $w > 0$ satisfies*

$$\text{Vol}(K) \geq \frac{130838246407123}{10^{15}}\pi w^3 > 0.411040473721188 w^3. \quad (1.3)$$

Together with (1.1), this gives the unconditional bracket

$$0.411040473721188 w^3 < m_3(w) \leq 0.419860045965080 \dots w^3.$$

The new lower bound is 7.94% larger than (1.2), and it is 97.8994% of the conjectured optimum.

The geometric mechanism is as follows. Normalize $w = 1$ and place the origin at the center of a minimum circumscribed ball of radius R . Constant width implies the opposite-point relation

$$x - n \in K$$

whenever $x \in \partial K$ and n is an outer unit normal at x . Since $K \subset B(0, R)$, this gives the pointwise estimate

$$\langle x, n \rangle \geq \frac{|x|^2 + 1 - R^2}{2}. \quad (1.4)$$

Applying the divergence theorem to

$$F_m(x) = \frac{2x}{|x|^2 + m}, \quad 0 < m \leq 1 - R^2,$$

yields, for the radial function ρ of K ,

$$S(K) \leq \int_{\mathbb{S}^2} \frac{2\rho(u)^3}{\rho(u)^2 + m} d\sigma(u), \quad 3 \text{Vol}(K) = \int_{\mathbb{S}^2} \rho(u)^3 d\sigma(u). \quad (1.5)$$

Blaschke's identity

$$S(K) = 2 \text{Vol}(K) + \frac{2\pi}{3} \quad (1.6)$$

then converts a suitable affine majorant of the integrand into a volume lower bound.

The affine majorant is tangent in the variable $X = \rho^3$. Its gap $G(\rho) \geq 0$ is discarded in the purely analytic argument, which already gives the following result.

Theorem 1.2 (Analytic bound). *Every convex body $K \subset \mathbb{R}^3$ of constant width $w > 0$ satisfies*

$$\text{Vol}(K) \geq \frac{533032}{4143735} \pi w^3 = 0.404120778796972 \dots w^3. \quad (1.7)$$

To prove theorem 1.1, we recover a positive portion of the discarded gap. When $R^2 > 1/3$, the minimum circumscribed ball has four contact directions $u_1, \dots, u_4$ whose mutual angles satisfy explicit two-sided bounds. The radial function is forced to be large in caps about the u_i by successive supporting-plane estimates, and small in caps about $-u_i$ because

$$K \subset B(Ru_i, 1).$$

The resulting eight caps are pairwise disjoint. Integrating $G(\rho)$ over rational spherical rings gives a finite exact-rational certificate. The circumradius interval

$$\frac{1}{2} \leq R \leq \frac{\sqrt{6}}{4}$$

is divided into finitely many subintervals, and every required inequality is verified with integer arithmetic.

The computer-assisted component is deliberately narrow. All geometric statements, all monotonicity arguments, and the reduction to finitely many rational inequalities are proved in the text. The remaining computation is an evaluation of explicit rational inequalities, carried out in exact rational arithmetic. Binary floating point is used only to locate a candidate grid point; each accepted point and every implication used in the proof are rechecked by exact rational inequalities.

Organization. Section 2 records the constant-width identities. The circum-sphere contact geometry and Jung's inequality are proved in section 3. Radial propagation estimates are developed in section 4. The divergence inequality is established in section 5. The tangent majorant and theorem 1.2 are proved in section 6. The eight-cap deficit estimate is given in section 7. The exact certificate and the optimized circumradius partition are described in section 8, and the main theorem is concluded in section 9. A certified rational lower bound for π and a purely algebraic form of the tangent gap are collected in the appendices.

2. CONSTANT-WIDTH IDENTITIES

Throughout the proof we normalize the width to be 1. The general statement follows by dilation. Let $B(z, r)$ denote the closed Euclidean ball of center z and radius r , and write $B = B(0, 1)$. Surface measure on $\mathbb{S}^2$ is denoted by σ , with $\sigma(\mathbb{S}^2) = 4\pi$.

For a convex body K , its support function is

$$h_K(u) = \max_{x \in K} \langle x, u \rangle, \quad u \in \mathbb{S}^2.$$

Lemma 2.1. *A convex body K has constant width 1 if and only if*

$$h_K(u) + h_K(-u) = 1 \quad (u \in \mathbb{S}^2).$$

Consequently,

$$K + (-K) = B, \quad \text{diam } K = 1. \quad (2.1)$$

Proof. The two supporting planes orthogonal to u have equations $\langle x, u \rangle = h_K(u)$ and $\langle x, u \rangle = -h_K(-u)$, so their distance is $h_K(u) + h_K(-u)$. Moreover,

$$h_{K+(-K)}(u) = h_K(u) + h_K(-u) = 1 = h_B(u),$$

which proves the first identity in (2.1). If $x, y \in K$ and $x \neq y$, then with $u = (x - y)/|x - y|$,

$$|x - y| = \langle x, u \rangle + \langle -y, u \rangle \leq h_K(u) + h_K(-u) = 1.$$

Conversely, two supporting planes at distance 1 contain points of K , so $\text{diam } K \geq 1$. $\square$

Lemma 2.2 (Blaschke's identity). *If $K \subset \mathbb{R}^3$ has constant width 1, volume V , and surface area S , then*

$$V = \frac{S}{2} - \frac{\pi}{3}, \quad \text{equivalently} \quad S = 2V + \frac{2\pi}{3}. \quad (2.2)$$

Proof. Let $V(\cdot, \cdot, \cdot)$ denote mixed volume, normalized by $V(A, A, A) = \text{Vol}(A)$. From (2.1) and multilinearity,

$$\frac{4\pi}{3} = \text{Vol}(K + (-K)) = 2V + 6V(K, K, -K).$$

On the other hand,

$$\frac{S}{3} = V(K, K, B) = V(K, K, K + (-K)) = V + V(K, K, -K).$$

Eliminating the mixed volume gives (2.2). We refer to [12, Chapter 5] for the standard mixed-volume facts used here. $\square$

3. THE MINIMUM CIRCUMSCRIBED BALL

Let $B(0, R)$ be a minimum-radius ball containing K , after translating its center to the origin, and set

$$\mathcal{F} = K \cap \partial B(0, R).$$

Lemma 3.1 (Contact balance). *The origin belongs to $\text{conv}(\mathcal{F})$. Consequently, there exist pairwise distinct directions $u_1, \dots, u_n \in \mathbb{S}^2$, with $2 \leq n \leq 4$, and positive numbers $\lambda_1, \dots, \lambda_n$ such that*

$$Ru_i \in K, \quad \sum_{i=1}^n \lambda_i = 1, \quad \sum_{i=1}^n \lambda_i u_i = 0. \quad (3.1)$$

Proof. Suppose $0 \notin \text{conv}(\mathcal{F})$. By strict separation there are a unit vector e and $\alpha > 0$ such that $\langle y, e \rangle \geq \alpha$ for every $y \in \mathcal{F}$. Put

$$A = \{y \in K : \langle y, e \rangle \leq \alpha/2\}.$$

If $A \neq \emptyset$, then A is compact and disjoint from $\mathcal{F}$, hence $\mu := \max_A |y| < R$. Choose $t > 0$ so small that $t < \alpha/2$ and, when $A \neq \emptyset$, $\mu + t < R$. If $y \notin A$, then

$$|y - te|^2 = |y|^2 - 2t\langle y, e \rangle + t^2 \leq R^2 - t\alpha + t^2 < R^2.$$

If $y \in A$, then $|y - te| \leq \mu + t < R$. Compactness now gives $K \subset B(te, R')$ for some $R' < R$, contradicting minimality. Thus $0 \in \text{conv}(\mathcal{F})$. Carathéodory's theorem gives a convex representation using at most four points; deleting zero coefficients gives (3.1). The case $n = 1$ is impossible. $\square$

Lemma 3.2. *For $i \neq j$,*

$$\langle u_i, u_j \rangle \geq c := 1 - \frac{1}{2R^2}. \quad (3.2)$$

Moreover $c < 0$. Writing

$$\eta := -c = \frac{1}{2R^2} - 1 > 0, \quad (3.3)$$

one has

$$\lambda_j \leq \frac{\eta}{1 + \eta}, \quad n \geq \frac{1 + \eta}{\eta} = \frac{1}{1 - 2R^2}. \quad (3.4)$$

If $n = 4$ and $\eta < 1$, then also

$$\langle u_i, u_j \rangle \leq t_{\max}(\eta) := \frac{4\eta^2}{1 - \eta} - 1 \quad (i \neq j). \quad (3.5)$$

Proof. By lemma 2.1,

$$1 \geq |Ru_i - Ru_j|^2 = 2R^2(1 - \langle u_i, u_j \rangle),$$

which proves (3.2). Taking the scalar product of (3.1) with u_j gives

$$0 = \lambda_j + \sum_{i \neq j} \lambda_i \langle u_i, u_j \rangle \geq \lambda_j + c(1 - \lambda_j).$$

Since $\lambda_j > 0$, this forces $c < 0$, and then yields $\lambda_j(1 + \eta) \leq \eta$. Summing over j proves (3.4).

Assume now $n = 4$. Fix the pair $(1, 2)$, set $t = \langle u_1, u_2 \rangle$ and $A = \lambda_1 + \lambda_2$. Applying the preceding scalar-product inequality for $j = 1, 2$ and adding gives

$$A(1 + t) \leq 2\eta(1 - A). \quad (3.6)$$

Applying it for $j = 3, 4$ and adding gives

$$1 - A \leq \eta(1 + A), \quad \text{hence} \quad A \geq \frac{1 - \eta}{1 + \eta}.$$

Substitution into (3.6) yields

$$1 + t \leq 2\eta \left(\frac{1}{A} - 1 \right) \leq \frac{4\eta^2}{1 - \eta},$$

which is (3.5). $\square$

Corollary 3.3 (Jung's inequality in dimension three). *The circumradius satisfies*

$$\frac{1}{2} \leq R \leq \frac{\sqrt{6}}{4}, \quad R^2 \leq \frac{3}{8}. \quad (3.7)$$

If $R^2 > 1/3$, then $n = 4$ in lemma 3.1 and

$$\frac{1}{3} \leq \eta < \frac{1}{2}. \quad (3.8)$$

Proof. The lower bound follows from $\text{diam } K = 1$ and $K \subset B(0, R)$. By (3.4) and $n \leq 4$,

$$\frac{1}{1 - 2R^2} \leq 4,$$

which is equivalent to $R^2 \leq 3/8$. If $R^2 > 1/3$, then (3.4) gives $n > 3$, so $n = 4$. Formula (3.3) gives (3.8). This is the classical theorem of Jung [6], here obtained directly from the contact balance. $\square$

Since $R < 1$, the origin lies in the interior of K . Indeed, $h_K(u) \leq R$ and constant width give $h_K(u) \geq 1 - R$, hence

$$B(0, 1 - R) \subset K \subset B(0, R). \quad (3.9)$$

We may therefore define the continuous radial function

$$\rho(u) = \max\{t \geq 0 : tu \in K\}, \quad u \in \mathbb{S}^2,$$

which satisfies

$$1 - R \leq \rho(u) \leq R. \quad (3.10)$$

For every contact direction in (3.1),

$$\rho(u_i) = R, \quad (3.11)$$

and the plane $\langle x, u_i \rangle = R$ supports K at Ru_i .

Lemma 3.4 (Ball bound). *Let $Ru \in K$, let $0 < R_* \leq R$, and let $v \in \mathbb{S}^2$ satisfy $c := -\langle v, u \rangle \in [0, 1]$. Then*

$$\rho(v)^2 + 2R_*c\rho(v) + R_*^2 \leq 1, \quad (3.12)$$

and hence

$$\rho(v) \leq \Pi(R_*, c) := -R_*c + \sqrt{1 - R_*^2(1 - c^2)}. \quad (3.13)$$

For fixed R_* , the function $c \mapsto \Pi(R_*, c)$ is strictly decreasing on $[0, 1]$, and $\Pi(R_*, 1) = 1 - R_*$.

Proof. The points $\rho(v)v$ and Ru belong to K , so their distance is at most 1. Thus

$$\rho(v)^2 + 2Rc\rho(v) + R^2 \leq 1.$$

Replacing R by the smaller R_* decreases the left-hand side and gives (3.12). Solving the quadratic inequality for its nonnegative root gives (3.13). Strict monotonicity follows either by differentiating or by comparing the quadratics $t^2 + 2R_*ct + R_*^2 - 1$ for different c . $\square$

4. OPPOSITE POINTS AND RADIAL PROPAGATION

Lemma 4.1 (Opposite point). *Let $x \in \partial K$ and let n be an outer unit normal of K at x . Then*

$$x - n \in K. \quad (4.1)$$

Consequently,

$$\langle x, n \rangle \geq \frac{|x|^2 + 1 - R^2}{2}. \quad (4.2)$$

Proof. Choose $y \in K$ on the opposite supporting plane. Since the width in direction n is 1,

$$\langle x - y, n \rangle = 1.$$

By lemma 2.1, $|x - y| \leq 1$. Equality in Cauchy–Schwarz therefore gives $x - y = n$, proving (4.1). Since $x - n \in K \subset B(0, R)$,

$$R^2 \geq |x - n|^2 = |x|^2 - 2\langle x, n \rangle + 1,$$

which is (4.2). $\square$

Fix from now on a number

$$\frac{5}{8} \leq m \leq 1 - R^2. \quad (4.3)$$

For $s \in [1 - R, R]$, define

$$\cos \gamma_m(s) = \frac{s^2 + m}{2s}, \quad 0 \leq \gamma_m(s) < \frac{\pi}{2}. \quad (4.4)$$

The expression lies in $(0, 1]$: the inequality $(s^2 + m)/(2s) \leq 1$ follows from $m \leq 1 - R^2$ and $s \in [1 - R, R]$. Moreover,

$$\frac{d}{ds} \frac{s^2 + m}{2s} = \frac{1}{2} \left(1 - \frac{m}{s^2} \right) < 0,$$

because $s^2 \leq R^2 \leq 3/8 < 5/8 \leq m$. Thus γ_m is strictly increasing. By (4.2), if $x = \rho(v)v$ and n is an outer normal at x , then

$$d(v, n) \leq \gamma_m(\rho(v)). \quad (4.5)$$

For $s \in [1 - R, R]$ and $0 \leq \delta < \pi/2$, define

$$N_m(s, \delta) := \frac{s \cos \gamma_m(s)}{\cos(\gamma_m(s) + \delta)} = \frac{s(s^2 + m)}{(s^2 + m) \cos \delta - \sqrt{4s^2 - (s^2 + m)^2} \sin \delta}, \quad (4.6)$$

whenever the denominator is positive.

Lemma 4.2 (Supporting-plane step). *Let $v, v' \in \mathbb{S}^2$, $d(v, v') = \delta$, and suppose $\gamma_m(\rho(v')) + \delta < \pi/2$. Then*

$$\rho(v) \leq N_m(\rho(v'), \delta). \quad (4.7)$$

For fixed admissible $\delta > 0$, $s \mapsto N_m(s, \delta)$ is strictly increasing; for fixed admissible s , $\delta \mapsto N_m(s, \delta)$ is nondecreasing.

Proof. Let $x = \rho(v')v'$ and choose an outer unit normal n at x . Set $\gamma = d(v', n)$. By (4.5), $\gamma \leq \gamma_m(\rho(v'))$. The supporting plane at x gives

$$\rho(v)\langle v, n \rangle \leq \langle x, n \rangle = \rho(v') \cos \gamma.$$

The spherical triangle inequality gives $d(v, n) \leq \delta + \gamma < \pi/2$, hence

$$\rho(v) \leq \rho(v') \frac{\cos \gamma}{\cos(\gamma + \delta)}.$$

The function $t \mapsto \cos t / \cos(t + \delta)$ is nondecreasing, since its derivative is $\sin \delta / \cos^2(t + \delta)$. This proves (4.7). The monotonicities follow from

$$N_m(s, \delta) = \frac{(s^2 + m)/2}{\cos(\gamma_m(s) + \delta)},$$

because the numerator and $\gamma_m(s)$ are strictly increasing in s . $\square$

Lemma 4.3 (Rational step certificate). *Let $X, Y, m, c_\delta, s_\delta$ be positive rationals with $c_\delta^2 + s_\delta^2 = 1$. Put*

$$T = X^2 + m, \quad D = 4X^2 - T^2.$$

Assume $D \geq 0$ and

$$T(Yc_\delta - X) > 0, \tag{4.8}$$

$$Y^2 D s_\delta^2 \leq T^2 (Yc_\delta - X)^2. \tag{4.9}$$

If $\delta \in (0, \pi/2)$ has cosine c_δ and sine s_δ , then

$$\gamma_m(X) + \delta < \frac{\pi}{2}, \quad N_m(X, \delta) \leq Y.$$

Proof. The left-hand side of (4.8) is positive. Taking square roots in (4.9) gives

$$Y\sqrt{D} s_\delta \leq T(Yc_\delta - X),$$

so

$$Y(Tc_\delta - \sqrt{D} s_\delta) \geq TX > 0.$$

The factor in parentheses equals $2X \cos(\gamma_m(X) + \delta)$ and is therefore positive. Dividing gives $Y \geq N_m(X, \delta)$. $\square$

Fix a positive integer N and define a rational angular grid by

$$\begin{aligned} t_k &= \frac{k}{N}, & \theta_k &= 2 \arctan t_k, \\ c_k &:= \cos \theta_k = \frac{1 - t_k^2}{1 + t_k^2}, & s_k &:= \sin \theta_k = \frac{2t_k}{1 + t_k^2}. \end{aligned} \tag{4.10}$$

For $k \geq 1$, set

$$c_k^\Delta = c_k c_{k-1} + s_k s_{k-1}, \quad s_k^\Delta = s_k c_{k-1} - c_k s_{k-1}. \tag{4.11}$$

These are exactly the cosine and sine of $\theta_k - \theta_{k-1}$, and all quantities are rational.

Proposition 4.4 (Vertex chain). *Let $u_0 \in \mathbb{S}^2$ satisfy $\rho(u_0) = R$, and let $R_* \leq R$. Suppose $\theta_{n_v} < \pi/2$ and*

$$w_0 = R_* > w_1 > \cdots > w_{n_v}$$

are rationals in $[1 - R, R]$, and that for each $1 \leq k \leq n_v$ the conditions (4.8)–(4.9) hold with

$$X = w_k, \quad Y = w_{k-1}, \quad c_\delta = c_k^\Delta, \quad s_\delta = s_k^\Delta.$$

Then

$$\rho(v) \geq w_k \quad \text{whenever} \quad d(v, u_0) \leq \theta_k. \quad (4.12)$$

Proof. Proceed by induction. The assertion at the center follows from $\rho(u_0) = R \geq R_*$. Suppose the statement holds for $k-1$, and let $d(v, u_0) \leq \theta_k$. If $d(v, u_0) \leq \theta_{k-1}$, there is nothing to prove. Otherwise, let v_* be the point at distance θ_{k-1} from u_0 on the minimizing geodesic to v , so $d(v, v_*) \leq \theta_k - \theta_{k-1}$. If $\rho(v) < w_k$, then lemmas 4.2 and 4.3 and the monotonicities of N_m imply

$$w_{k-1} \leq \rho(v_*) \leq N_m(\rho(v), d(v, v_*)) < N_m(w_k, \theta_k - \theta_{k-1}) \leq w_{k-1},$$

a contradiction. $\square$

Proposition 4.5 (Antipodal cap bound). *Let $Ru_0 \in K$, let $0 < R_* \leq R$, let k satisfy $\theta_k < \pi/2$, and suppose a rational z_k satisfies*

$$z_k + R_* c_k \geq 0, \quad (4.13)$$

$$(z_k + R_* c_k)^2 \geq 1 - R_*^2 s_k^2. \quad (4.14)$$

Then

$$\rho(v) \leq z_k \quad \text{whenever} \quad d(v, -u_0) \leq \theta_k. \quad (4.15)$$

Proof. Conditions (4.13)–(4.14) are equivalent to

$$z_k \geq \Pi(R_*, c_k).$$

If $d(v, -u_0) \leq \theta_k$, then $-\langle v, u_0 \rangle \geq c_k$, and lemma 3.4 gives

$$\rho(v) \leq \Pi(R_*, -\langle v, u_0 \rangle) \leq \Pi(R_*, c_k) \leq z_k.$$

$\square$

5. A DIVERGENCE INEQUALITY

We first prove the estimate for bodies with C^1 boundary; the general case follows by outer parallel regularization.

Proposition 5.1 (Transport inequality). *Let K have constant width 1, let the origin be its circumcenter, and assume ∂K is C^1 . If $0 < m \leq 1 - R^2$, then*

$$3V = \int_{\mathbb{S}^2} \rho(u)^3 \, d\sigma(u), \quad (5.1)$$

$$S \leq \int_{\mathbb{S}^2} \frac{2\rho(u)^3}{\rho(u)^2 + m} \, d\sigma(u). \quad (5.2)$$

Proof. The first identity is spherical integration over the star-shaped set K .
For the second, define

$$F_m(x) = \frac{2x}{|x|^2 + m}.$$

By (4.2), at every $x \in \partial K$ with outer unit normal n ,

$$\langle F_m(x), n \rangle = \frac{2\langle x, n \rangle}{|x|^2 + m} \geq 1.$$

The divergence theorem therefore gives

$$S \leq \int_{\partial K} \langle F_m, n \rangle \, d\mathcal{H}^2 = \int_K \operatorname{div} F_m \, dx.$$

Since

$$\operatorname{div} F_m(x) = \frac{2(|x|^2 + 3m)}{(|x|^2 + m)^2}$$

and

$$\frac{d}{dr} \left(\frac{2r^3}{r^2 + m} \right) = r^2 \frac{2(r^2 + 3m)}{(r^2 + m)^2},$$

spherical integration gives (5.2). □

Lemma 5.2 (Regularization). *For $\varepsilon > 0$, define*

$$K_\varepsilon = \frac{K + \varepsilon B}{1 + 2\varepsilon}.$$

Then K_ε has constant width 1, its boundary is C^1 , and $\operatorname{Vol}(K_\varepsilon) \rightarrow \operatorname{Vol}(K)$ as $\varepsilon \downarrow 0$.

Proof. The support function satisfies

$$h_{K_\varepsilon}(u) = \frac{h_K(u) + \varepsilon}{1 + 2\varepsilon},$$

so $h_{K_\varepsilon}(u) + h_{K_\varepsilon}(-u) = 1$. The outer parallel body $K + \varepsilon B$ has a unique outer normal at every boundary point: the metric projection onto the closed convex set K is unique, and the outer normal is the normalized displacement from the nearest point. Locally the boundary is the graph of an everywhere differentiable convex function, hence is C^1 ; see [11, Theorem 25.5]. Finally, Hausdorff convergence of $K + \varepsilon B$ to K implies convergence of volume; see [12]. □

Consequently, any uniform lower bound proved using proposition 5.1 passes to arbitrary convex bodies of constant width.

6. THE TANGENT MAJORANT AND AN ANALYTIC BOUND

For $m > 0$, define

$$\phi_m(X) = \frac{2X}{X^{2/3} + m}, \quad X > 0.$$

A direct computation gives

$$\phi_m''(X) = -\frac{4(X^{2/3} + 5m)}{9X^{1/3}(X^{2/3} + m)^3} < 0. \quad (6.1)$$

Thus ϕ_m is strictly concave.

Fix a rational tangency radius $s_0 = p/q > 0$ and set

$$a = \phi_m'(s_0^3) = \frac{2(s_0^2 + 3m)}{3(s_0^2 + m)^2}, \quad b = \phi_m(s_0^3) - as_0^3. \quad (6.2)$$

Define the tangent gap

$$G_{m,s_0}(s) = as^3 + b - \frac{2s^3}{s^2 + m}. \quad (6.3)$$

Lemma 6.1. *For $s > 0$,*

$$G_{m,s_0}(s) \geq 0,$$

with equality if and only if $s = s_0$. Moreover, G_{m,s_0} is strictly decreasing on $(0, s_0]$ and strictly increasing on $[s_0, \infty)$.

Proof. In the variable $X = s^3$, the function $aX + b$ is the tangent line to the strictly concave function ϕ_m at $X = s_0^3$, hence lies above it. Since ϕ_m' is strictly decreasing, the derivative of the tangent gap has the sign of $X - s_0^3$, proving the monotonicity. $\square$

Let

$$E = \int_{\mathbb{S}^2} G_{m,s_0}(\rho(u)) \, d\sigma(u) \geq 0. \quad (6.4)$$

Proposition 6.2 (Master inequality). *Assume $m \leq 1 - R^2$ and $3a - 2 > 0$. Then*

$$V \geq \frac{2\pi/3 - 4\pi b + E}{3a - 2}. \quad (6.5)$$

Proof. By (6.3), (5.1), and (5.2),

$$S \leq \int_{\mathbb{S}^2} (a\rho^3 + b - G_{m,s_0}(\rho)) \, d\sigma = 3aV + 4\pi b - E.$$

Using $S = 2V + 2\pi/3$ and rearranging proves (6.5). $\square$

Proof of theorem 1.2. Normalize $w = 1$. By corollary 3.3, one may take $m = 5/8$. Choose $s_0 = 11/24$. Exact arithmetic gives

$$a = \frac{461184}{231361}, \quad b = \frac{161051}{4164498}, \quad 3a - 2 = \frac{920830}{231361} > 0.$$

Discarding E in (6.5),

$$V \geq \frac{(2/3 - 4b)\pi}{3a - 2} = \frac{533032}{4143735}\pi.$$

Apply lemma 5.2 and then rescale to width w . $\square$

7. RECOVERING THE DEFICIT ON EIGHT SPHERICAL CAPS

Fix numbers

$$R_* \leq R \leq R^+, \quad R_*^2 > \frac{1}{3}, \quad m := 1 - (R^+)^2 \geq \frac{5}{8}, \quad (7.1)$$

where R_* is rational and m is rational. In the top interval we allow $R^+ = \sqrt{6}/4$, so $m = 5/8$. Fix a rational tangency radius $s_0 \in (1 - R_*, R_*)$ and write $G = G_{m, s_0}$.

By corollary 3.3, there are exactly four contact directions $u_1, \dots, u_4$. Put

$$\eta_* = \frac{1}{2R_*^2} - 1, \quad T_* = \frac{4\eta_*^2}{1 - \eta_*} - 1. \quad (7.2)$$

Since $R \geq R_*$, the actual η is at most η_* , and $t_{\max}(\eta)$ is increasing for $0 < \eta < 1$. Hence

$$\langle u_i, u_j \rangle \leq T_* \quad (i \neq j). \quad (7.3)$$

Let $w_0 = R_* > w_1 > \dots > w_{n_v} > s_0$ satisfy the exact step certificates of lemma 4.3, and let $z_1, \dots, z_{n_a} \leq s_0$ satisfy the exact ball certificates (4.13)–(4.14). Define

$$C_i^+ = \{v \in \mathbb{S}^2 : d(v, u_i) \leq \theta_{n_v}\}, \quad C_i^- = \{v \in \mathbb{S}^2 : d(v, -u_i) \leq \theta_{n_a}\}.$$

Lemma 7.1 (Disjointness). *Assume $\theta_{n_v}, \theta_{n_a} < \pi/2$ and*

$$\cos(2\theta_{n_v}) > T_*, \quad \cos(2\theta_{n_a}) > T_*. \quad (7.4)$$

Then the eight caps $C_i^\pm$ are pairwise disjoint.

Proof. For $i \neq j$, (7.3) and the first inequality in (7.4) imply $d(u_i, u_j) > 2\theta_{n_v}$, so the four positive caps are pairwise disjoint. The same argument applies to the four negative caps.

If $v \in C_i^+$, then proposition 4.4 gives $\rho(v) \geq w_{n_v} > s_0$. If $v \in C_j^-$, then proposition 4.5 gives $\rho(v) \leq z_{n_a} \leq s_0$. Thus a positive and a negative cap cannot intersect, even when $i \neq j$. $\square$

The area of the spherical ring $\theta_{k-1} \leq d(v, u) \leq \theta_k$ is $2\pi(c_{k-1} - c_k)$. Using the monotonicity of G on either side of s_0 yields the following quantitative estimate.

Proposition 7.2 (Eight-cap deficit). *Under the hypotheses above, define*

$$\Delta = 4 \left[\sum_{k=1}^{n_v} (c_{k-1} - c_k) G(w_k) + \sum_{k=1}^{n_a} (c_{k-1} - c_k) G(z_k) \right]. \quad (7.5)$$

Then

$$E \geq 2\pi\Delta, \quad (7.6)$$

and consequently

$$V \geq \pi \frac{2/3 - 4b + 2\Delta}{3a - 2}. \quad (7.7)$$

Proof. On the k th ring about u_i , proposition 4.4 gives $\rho \geq w_k > s_0$, so lemma 6.1 gives $G(\rho) \geq G(w_k)$. On the k th ring about $-u_i$, proposition 4.5 gives $\rho \leq z_k \leq s_0$, and the decreasing branch of G gives $G(\rho) \geq G(z_k)$. Integrate over the rings and sum over the eight disjoint caps. This proves (7.6); substituting in (6.5) gives (7.7). $\square$

8. THE EXACT RATIONAL CERTIFICATE

This section reduces the remaining proof to a finite exact computation. Every quantity below is an explicit rational number, and every comparison is an integer comparison, so the verification requires only exact rational arithmetic.

8.1. What is verified. For each circumradius interval $[R_*, R^+]$ in tables 1 and 2, the verification performs the following steps.

Verification protocol 8.1.

- (i) Set $m = 1 - (R^+)^2$, with $m = 5/8$ in the top interval, and set the rational tangency radius $s_0 = p/q$ listed in the table. Compute a, b from (6.2) and verify $3a - 2 > 0$.
- (ii) Form the rational angles (4.10)–(4.11) with the listed N .
- (iii) Construct a decreasing sequence $w_0 = R_* > w_1 > \cdots > w_{n_v} > s_0$ on the grid $D^{-1}\mathbb{Z}$. Every accepted step is rechecked by the exact inequalities (4.8)–(4.9).
- (iv) Construct upper bounds $z_k \in D^{-1}\mathbb{Z}$ for the antipodal caps. Every accepted value is rechecked by the exact inequalities (4.13)–(4.14), and the retained sequence satisfies $z_k \leq s_0$.
- (v) Verify the range and monotonicity conditions, $n_v, n_a < N$ (hence $\theta_{n_v}, \theta_{n_a} < \pi/2$), $w_{n_v} > s_0 \geq z_{n_a}$, and the two exact disjointness inequalities (7.4).
- (vi) Evaluate the two sums in (7.5) using exact fractions. After each addition, round downward to a multiple of 10^{-18} ; this can only decrease the certified deficit. Evaluate the right-hand side of (7.7) and round its coefficient of π downward to a multiple of 10^{-15} .

Floating-point arithmetic appears only in the search for nearby candidate grid integers in steps (iii) and (iv). It is not used in any comparison supporting the theorem. A candidate that is inaccurate is moved until the corresponding exact inequalities pass; a candidate that is unnecessarily conservative merely weakens the bound.

The interval $R \leq 3/5$ needs no cap certificate. There one takes $m = 16/25$, $s_0 = 37/80$, and $E \geq 0$ in (6.5), obtaining

$$V \geq \frac{76439431}{575385750} \pi = 0.417357494296971 \dots \quad (8.1)$$

8.2. Certified partition. The following two tables give the complete gap-free partition of $[1/2, \sqrt{6}/4]$. The displayed final volume bounds are decimal renderings only; the verification stores and compares exact rational coefficients of π . Here D is the common denominator of the w_k and z_k grids.

TABLE 1. Coarser part of the exact certificate. All endpoints and tangency radii are rational.

Case	$[R_*, R^+]$	s_0	N	D	n_v	n_a	certified V
0	$[1/2, 3/5]$	37/80	—	—	—	—	0.417357494296971
O1	$[3/5, 3047/5000]$	59/125	38400	10^9	7922	14954	0.411496539437612
O2	$[3047/5000, 30549/50000]$	61/125	38400	10^9	6981	17485	0.411755637432267
O3	$[30549/50000, 24469/40000]$	123/250	38400	10^9	6723	17997	0.411346798844895
O4	$[24469/40000, 6121/10000]$	247/500	38400	10^9	6595	18245	0.411138407971101
O5	$[6121/10000, 24491/40000]$	249/500	38400	10^9	6315	18630	0.411062159522339
O6	$[24491/40000, 244937/400000]$	1/2	38400	10^9	6177	18819	0.411041517359710
A1	$[244937/400000, 15309/25000]$	$5001359/10^7$	38400	10^9	6170	18837	0.411043738157074
A2	$[15309/25000, 122473/200000]$	$500213/10^6$	38400	10^9	6165	18845	0.411043774241325
A3	$[122473/200000, 38273/62500]$	$500231/10^6$	38400	10^9	6164	18847	0.411041898741644
A4	$[38273/62500, 1224739/2000000]$	250121/500000	38400	10^9	6163	18848	0.411041134307505

TABLE 2. Refined part of the exact certificate near the Jung endpoint.

Case	$[R_*, R^+]$	s_0	N	D	n_v	n_a	certified V
B1	$[1224739/2000000, 1224741/2000000]$	$5002481/10^7$	76800	10^9	12326	37698	0.411041098837742
C1	$[1224741/2000000, 1224743/2000000]$	1000501/2000000	153600	10^9	24651	75397	0.411040501761867
C2	$[1224743/2000000, 3061859/5000000]$	2501279/5000000	153600	10^9	24650	75399	0.411040483688246
D1	$[3061859/5000000, 153093/250000]$	15633/31250	307200	10^{12}	49306	150800	0.411040702618581
D2	$[153093/250000, 3061861/5000000]$	1000513/2000000	307200	10^{12}	49306	150800	0.411040531599784
D3	$[3061861/5000000, 6123723/10^7]$	1250643/2500000	307200	10^{12}	49306	150801	0.411040480663395
E1	$[6123723/10^7, 1530931/2500000]$	200103/400000	614400	10^{13}	98612	301603	0.411040482016165
E2	$[1530931/2500000, \sqrt{6}/4]$	25013/50000	614400	10^{13}	98609	301607	0.411040473721189

The exact coefficient of π in the last row is

$$\frac{130838246407123}{10^{15}}. \quad (8.2)$$

Every other row has a strictly larger certified coefficient. The endpoints in the tables match exactly, so the listed cases cover the entire Jung interval without gaps.

9. ASSEMBLY OF THE PROOF

Proof of theorem 1.1. Normalize $w = 1$ and apply lemma 5.2; it is enough to prove the bound for a C^1 body. Let R be the circumradius. By corollary 3.3,

$$\frac{1}{2} \leq R \leq \frac{\sqrt{6}}{4}.$$

If $R \leq 3/5$, (8.1) applies. Otherwise R lies in exactly one of the nonzero intervals of tables 1 and 2. The exact verification described in algorithm 8.1 establishes every hypothesis of proposition 7.2 on that interval and therefore

proves the corresponding row bound. The smallest exact coefficient is (8.2), attained in case E2. Hence

$$V \geq \frac{130838246407123}{10^{15}}\pi.$$

The certified decimal inequality is proved in appendix A. Finally, dilation by w multiplies volume by w^3 . $\square$

Remark 9.1 (Scope of the result). The theorem does not identify a minimizer and does not prove the Meissner conjecture. It narrows the universal interval to

$$0.411040473721188 < \frac{m_3(w)}{w^3} \leq 0.419860045965080 \dots$$

The difficult branch is the near-Jung regime $R \approx \sqrt{6}/4$, where the four circumsphere contact directions are forced to be nearly tetrahedral. This structural localization is independent of any assumption that an extremizer has Meissner geometry.

APPENDIX A. A CERTIFIED RATIONAL LOWER BOUND FOR π

The Taylor series of $\arcsin$ at $1/2$ has positive terms:

$$\pi = 6 \arcsin \frac{1}{2} = 6 \sum_{k=0}^{\infty} \frac{\binom{2k}{k}}{(2k+1)2^{4k+1}}.$$

Therefore the 40-term partial sum

$$\pi_{40} := 6 \sum_{k=0}^{39} \frac{\binom{2k}{k}}{(2k+1)2^{4k+1}} < \pi \tag{A.1}$$

provides an exact rational lower bound. Reducing this fraction to lowest terms, the following comparison holds by integer cross-multiplication:

$$\frac{130838246407123}{10^{15}}\pi_{40} > \frac{411040473721188}{10^{15}}.$$

Together with (A.1), this proves the strict decimal inequality in (1.3) without relying on a floating-point value of π .

APPENDIX B. A PURELY ALGEBRAIC FORM OF THE TANGENT GAP

Although concavity gives the shortest proof of lemma 6.1, the gap also admits a polynomial factorization, which gives an independent exact proof. For $s_0 = p/q$ with positive integers $p < q$, define

$$P(s) = (8p^2q + 24mq^3)s^3 + (16p^3 + 48mpq^2)s^2 + 32mp^2qs + 16mp^3.$$

Then

$$\begin{aligned} 12q^5(s_0^2 + m)^2 \left[(as^3 + b)(s^2 + m) - 2s^3 \right] \\ = (qs - p)^2 P(s). \end{aligned}$$

All coefficients of P are positive, so the factorization gives a second exact proof that $G(s) \geq 0$ for $s \geq 0$. Differentiating and simplifying yields

$$\begin{aligned} & 2qP(s)(s^2 + m) + (qs - p) \left(P'(s)(s^2 + m) - 2sP(s) \right) \\ &= 24qs^2(p + qs) \left(5m^2q^2 + 3mp^2 + (3mq^2 + p^2)s^2 \right), \end{aligned}$$

which certifies the monotonicity of the gap using rational polynomial identities alone.

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